

Vol 13 Chapter 4 — DNA and Proton Siblings

Keeper (author pass — deep math/physics revision)

2026-05-24 Sunday

Chapter 4 — DNA and Proton Siblings

Level 1 — one sentence

DNA's double-helix architecture — 2 strands (= rank), antiparallel orientation, Watson-Crick base pairing via hydrogen bonds (proton sharing), ≈ 10.5 bp/turn major groove — exhibits BST-primary structural signatures, and the proton-mediated H-bond pairing reveals DNA as a substrate K-type with proton as substrate-mediator.

Level 2 — graduate-physicist precision

4.1 DNA architecture

- Double helix: 2 antiparallel strands (5'-3' opposite)
- Rank = 2 strands
- Diameter: 2 nm
- Major groove width: 2.2 nm; minor groove: 1.2 nm
- 10.5 bp / helix turn (B-form)
- Watson-Crick base pairs: A-T (2 H-bonds), G-C (3 H-bonds)

Average G+C content varies by organism: $\sim 30\text{-}70\%$.

4.2 Hydrogen bonds in DNA

A-T pair: 2 H-bonds G-C pair: 3 H-bonds (one extra)

H-bond energy $\sim 10\text{-}40$ kJ/mol (Vol 12 Ch 2). DNA stability comes from cumulative H-bonding + stacking (van der Waals between adjacent bases).

Melting temperature T_m : increases with G+C fraction ($\sim 0.4^\circ\text{C}$ per % G+C).

4.3 Proton siblings

Each H-bond shares one proton between donor and acceptor.

BST view: H-bond = substrate K-type configuration with proton playing substrate-mediator role. Proton substrate K-type (Vol 5 Ch 6: hydrogen atom; Vol 3 Ch 4: nucleon-level) makes the DNA pairing.

The proton's substrate role here is non-trivial: $m_p/m_e = 6\pi^5$ (T1426; Vol 3) sets the proton's substrate K-type. The proton's BST-primary mass ratio enters DNA stability indirectly.

4.4 Superhelical structure — RETRACTED per Toy 3518 (2026-05-24)

HONEST RETRACTION (Elie Toy 3518, Sunday 2026-05-24): the “ ~ 137 base pairs per topological repeat” claim is RETRACTED. Canonical nucleosomal repeat is ~ 147 bp (147 bp wrapped around the histone octamer + ~ 50 bp

linker = ~200 bp full nucleosomal repeat). There is no documented ~137 bp superhelical structure in standard DNA topology literature.

$N_{\max} = 137$ does NOT appear at chromatin scale in any verifiable structural feature. The earlier “may be load-bearing” speculation was unsupported. Substrate’s $N_{\max} = 137$ ceiling remains operative at the atomic/element scale (Vol 12 Ch 1) but does not extend to DNA-scale topology via this claim.

This retraction is preserved per Quaker discipline (Vol 15 Ch 7): honest negatives strengthen the framework. The chapter’s other content (rank=2 strands, H-bonds, proton-mediator role, RS-coding parallel) is unaffected.

4.5 Information density

DNA stores 2 bits per base pair (4 states). Bacterial genome $\sim 10^6$ bp = ~2 Mbits. Human genome $\sim 3 \cdot 10^9$ bp = ~6 Gbits.

Information density: $\sim 10^{21}$ bits/cm³ (DNA) — competitive with technological storage.

Substrate view (Paper #122): DNA implements substrate’s Reed-Solomon code (Vol 14 Ch 9) at biological scale.

4.6 K-audit anchors

- Vol 5 Ch 6 (hydrogen atom)
- Vol 3 Ch 4 (proton structure)
- T1426 ($m_p/m_e = 6\pi^5$)

Level 3 — 5th-grader accessibility

DNA double helix: 2 strands (= rank), held by hydrogen bonds. A pairs with T (2 H-bonds), G with C (3). Each H-bond shares one proton. In BST: H-bonds are substrate K-types with proton-mediator role. The proton’s $m_p/m_e = 6\pi^5$ (Vol 3) enters DNA stability indirectly. Superhelical 137-bp repeats: RETRACTED — canonical nucleosomal repeat is ~147 bp, not 137 (Elie Toy 3518, Sunday 2026-05-24); substrate $N_{\max}$ doesn’t extend to chromatin scale via this claim. Information density: $\sim 10^{21}$ bits/cm³.

What comes next

Chapter 5 develops evolution as substrate dynamics.

Where to look this up

- Watson-Crick 1953
- BST: Vol 5 Ch 6; Vol 3 Ch 4; Paper #122